

torch.lerp#

<https://pytorch.org/docs/stable/generated/torch.lerp.html#torch.lerp>

Description:

Does a linear interpolation of two tensors start (given by input) and end based on a scalar or tensor weight and returns the resulting out tensor. The shapes of start and end must be broadcastable. If weight is a tensor, then the shapes of weight, start, and end must be broadcastable. input (Tensor) – the tensor with the starting points end (Tensor) – the tensor with the ending points weight (float or tensor) – the weight for the interpolation formula

Function Signature

```
torch.lerp(input, end, weight, *, out=None)#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> start = torch.arange(1., 5.)
>>> end = torch.empty(4).fill_(10)
>>> start
tensor([ 1.,  2.,  3.,  4.])
>>> end
tensor([ 10.,  10.,  10.,  10.])
>>> torch.lerp(start, end, 0.5)
tensor([ 5.5000,  6.0000,  6.5000,  7.0000])
>>> torch.lerp(start, end, torch.full_like(start, 0.5))
tensor([ 5.5000,  6.0000,  6.5000,  7.0000])
```

Detailed Documentation

Documentation

Does a linear interpolation of two tensors start (given by input) and end based on a scalar or tensor weight and returns the resulting out tensor.

The shapes of start and end must be broadcastable. If weight is a tensor, then the shapes of weight, start, and end must be broadcastable.

input (Tensor) – the tensor with the starting points

end (Tensor) – the tensor with the ending points

weight (float or tensor) – the weight for the interpolation formula

out (Tensor, optional) – the output tensor.